

Prasanna Kailash Warad

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SUMMARY

Data and AI Engineer with production experience designing ETL/ELT pipelines, LLM-orchestrated workflows, and ML systems that eliminate manual bottlenecks and drive operational decisions at scale. Engineered an agentic AI screening platform, collapsing a 5-stage hiring process to under 2 minutes and reducing reporting latency 28% across 2M+ retail transactions at Dollar General (Fortune 100). Core stack: Python, SQL, Apache Spark, Airflow, LLM orchestration, agentic AI (n8n), PostgreSQL, FastAPI, Docker, and AWS.

SKILLS

Agentic AI & LLM: n8n, Groq API, Ollama, LLaMA 3.3-70b, LLM Orchestration, RAG, Prompt Engineering, Whisper (OpenAI), Pinecone
Data Engineering: ETL/ELT, Apache Spark, Apache Airflow, Apache Kafka, Hadoop (HDFS/MapReduce), dbt, Databricks, Data Modeling
Programming: Python, TypeScript, SQL (Advanced), PySpark, FastAPI, Flask, Node.js, Express.js, Pandas, NumPy, scikit-learn, PyTorch, XGBoost, R
Computer Vision & AI Services: OpenCV, DeepFace, MediaPipe
Databases & Cloud: PostgreSQL, Redis, MySQL, MongoDB, Snowflake, Microsoft SQL Server, AWS (S3, Redshift, CloudWatch), Azure Synapse Analytics
Tools & Platforms: Docker, Terraform, Git, GitHub, CI/CD, REST APIs, Tableau, Power BI

EXPERIENCE

Cloud BC Labs, Reston, Virginia: Software Engineer Intern — Data & AI Engineering Oct 2025 – Present

- Automated** the full candidate assessment lifecycle using n8n agentic workflow and Groq LLM; job description submission triggers role-specific question generation and candidate email delivery with no manual steps, collapsing a 5-stage hiring process to under 2 minutes end-to-end.
- Designed** 6 PostgreSQL schema migrations from scratch — normalizing candidates, assessments, AI scores, proctoring flags, and job requisitions into a structured schema powering real-time recruiter dashboard queries for verdict tracking and pipeline reporting.
- Integrated** 4 AI services — DeepFace for identity verification, Groq LLM for MCQ scoring, Groq Whisper for video transcription, and MediaPipe for behavioral proctoring — into a 6-container Docker pipeline with no manual routing between stages.
- Deployed** RESTful APIs in FastAPI and Node.js, connecting the React HR dashboard to backend AI scoring services; standardized data contracts across 4 microservices, cutting cross-service integration overhead by 20%.
- Instrumented** an end-to-end screening funnel with usage analytics; surfaced recurring failure patterns that cut incoming bug report volume 15% and shifted sprint priorities toward the highest-impact fixes.

HCLTech Ltd, Noida, India: Data Engineer — Dollar General (Fortune 100 Retailer) Feb 2023 – Jul 2024

- Implemented** Spark and Airflow data pipelines with partition pruning and query caching, cutting operational reporting latency 28% across Dollar General's retail analytics stack serving 19,000+ store locations.
- Validated** 2M+ retail transaction records via Python and SQL anomaly detection, resolving data quality gaps that were degrading inventory replenishment model accuracy across Dollar General's fulfillment network.
- Constructed** an anomaly detection and downtime forecasting model using scikit-learn and PyTorch on AWS system performance logs, cutting incident response time by 25% by flagging at-risk systems before failures reached production.

PROJECTS [\(prasannawarad.com/projects\)](https://prasannawarad.com/projects)

Credit Risk Modeling & Default Prediction Pipeline | Python · Scikit-learn · XGBoost · SMOTE

- Applied** SMOTE to correct a 1:7.6 class imbalance across 255K+ lending records, then benchmarked logistic regression, random forest, and XGBoost via 5-fold cross-validation—XGBoost topped the comparison at 88.5% accuracy and 0.737 ROC-AUC.
- Compared** three classifiers by engineering 6 predictors and measuring precision-recall tradeoffs across each model, proposing a hybrid deployment strategy designed to catch high-risk defaults without over-flagging creditworthy borrowers in production.

Scalable Fleet Risk Analytics Pipeline | Hadoop · HDFS · MapReduce · Power BI

- Architected** a distributed data ingestion pipeline using Hadoop HDFS and MapReduce to process operational datasets across 300+ assets in 14 states, reducing manual data prep time and enabling repeatable pipeline runs across any fleet expansion.
- Developed** Power BI dashboards on the cleaned 300+ asset fleet dataset, surfacing risk scores and maintenance flags, enabling operations teams to shift from reactive incident response to predictive maintenance scheduling.

Instacart Customer Behavior Dashboard | SQL · Tableau · Python

- Processed** 3M+ grocery transaction records through optimized SQL pipelines—developing customer segmentation logic, cohort retention queries, and demand forecasting models that fed directly into operational reporting for merchandising and fulfillment planning teams.
- Produced** Tableau dashboards tracking supply chain KPIs, including product demand velocity, fulfillment rate trends, and campaign ROI—giving merchandising and operations teams a shared view for inventory planning and replenishment decisions.

CardioRisk — Predictive Health Analytics Pipeline | R · Statistical Modeling · ML Pipeline

- Evaluated** 5 classification models on 319K+ CDC health records—Neural Network achieved top performance at 91.4% accuracy, 0.81 AUC, and 99.97% recall; surfaced Decision Tree overfitting by flagging its matching accuracy against a 0.50 AUC, a critical signal for clinical deployment readiness.
- Ranked** BMI, age category, sleep time, and general health as top predictors from 17 clinical attributes via Random Forest Gini importance—adjusted classification threshold to maximize recall and minimize missed diagnoses.

EDUCATION

The University of Texas at Dallas, Texas (GPA: 3.88/4.0)	USA
M.S., Information Technology & Management — Dean's Excellence Scholar • Grad Certificate in Business Analytics & Data Mining	Aug 2024 – May 2026
Savitribai Phule Pune University, Pune (GPA: 3.80/4.0)	India
B.E., Electrical Engineering	Aug 2018 – May 2022